SOPHIA YOO

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Research Interests

Network Security & Privacy | Software-Defined Networking | Programmable Data Planes | HW/SW Co-Design

Academic Appointments

Assistant Professor of Computer Science, Amherst College 2026 – Present

Education

Princeton University, Princeton, NJ

- Ph.D. in Electrical and Computer Engineering, 2026
  - Dissertation: *Cooperative Network Systems for Protecting Internet Users*
  - Committee: Jennifer Rexford (Advisor), Maria Apostolaki (Advisor), David Hay, Prateek Mittal, Adrian Perrig
- M.A. in Electrical and Computer Engineering, 2022
  - Committee: Jennifer Rexford (Advisor), Prateek Mittal, Ravi Netravali

Temple University, Philadelphia, PA

- B.S. in Electrical Engineering, *summa cum laude*, 2020
  - Computer Engineering Concentration
  - Minors in Computer Science and Music

Publications

Sophia Yoo, Xiaoqi Chen, Jennifer Rexford. "SmartCookie: Blocking Large-Scale SYN Floods with a Split-Proxy Defense on Programmable Data Planes." *33rd USENIX Security Symposium.* (SEC'24).

Sophia Yoo*, Jiarong Xing*, Xenofon Foukas, Daehyeok Kim, Michael K. Reiter. "On the Criticality of Integrity Protection in 5G Fronthaul Networks." *33rd USENIX Security Symposium.* (SEC'24).

Henry Birge-Lee, **Sophia Yoo**, Benjamin Herber, Jennifer Rexford, Maria Apostolaki. "Tango: Secure Collaborative Route Control across the Public Internet." *21st USENIX Symposium on Networked Systems Design and Implementation.* (NSDI'24).

Sophia Yoo, Satadal Sengupta, Maria Apostolaki, Jennifer Rexford. "Sieve: Layered Network Defenses against Large-Scale Attacks." *Open Networking Foundation 2023 P4 Workshop.* (P4'23).

Sophia Yoo, Xiaoqi Chen. "Secure Keyed Hashing on Programmable Switches." ACM SIGCOMM 2021 Workshop on Secure Programmable Network Infrastructure. (SPIN'21).

Patents

Sophia Yoo*, Jiarong Xing*, Xenofon Foukas, Daehyeok Kim. “Security in 5G Fronthaul Networks.” US Patent Application No. 18/743,792, filed June 2024. Patent Pending.

Sophia Yoo, Xiaoqi Chen, Jennifer Rexford. “Split-layer Defense Against Volumetric Attacks with Programmable Data Planes.” Patent Pending.

Selected Awards and Honors

Finalist, Three-Minute Thesis (3MT) Competition, Princeton University (2026)

Wallace Memorial Fellowship, Highest Graduate Award in Engineering Research, Princeton University (2025)

Applied Networking Research Prize (ANRP) for *Tango [NSDI’2024]*, Internet Research Task Force (2025)

Graduate Student Award for Excellence in Service, ECE Department, Princeton University (2023-2024)

Graduate Research Fellowship Program Award (DGE-2039656), National Science Foundation (2022-2026)

Pramod Subramanian *17 Early Career Graduate Award, Princeton University (2022)

Princeton Research Day Innovation & Entrepreneurial Mindset Award, Princeton University (2022)

Diamond Award, Highest Undergraduate Academic Recognition, Temple University (2020)

Top Honors Recognition, ECE Department, Temple University (2020)

President's Scholar Award, Full-Tuition Merit Scholarship, Temple University (2016-2020)

Teaching Experience

Amherst College, Department of Computer Science - Instructor of Record

- Networks (COSC-283): Fall 2026
- Systems II - Operating Systems (COSC-275): Fall 2026, Spring 2027
- Research Seminar in Computer Networks: Spring 2027

Princeton University, Department of Electrical and Computer Engineering

- Course Development Contributor, Computer Networks, Fall 2025
- Assistant Instructor, Information Security: Spring 2022

Temple University, Department of Electrical and Computer Engineering

- Diamond Peer Teacher, Principles of Electric Circuits: Spring 2020
- Undergraduate Teaching Assistant, Intro to Engineering: Spring 2018, Fall 2018, Spring 2019
- STEM Tutor, Student Success Center: Fall 2017, Spring 2018, Fall 2018, Spring 2019, Fall 2019

Mentorship & Advising Experience

Veronika Kitsul, Princeton University (Fall 2025 - Spring 2026)
- Senior thesis research: “Peeling Back the Layers: Application- and Network-Level Characterization of ChatGPT”
- Continued to graduate study at the University of Michigan
- Co-advised with Jennifer Rexford, Satadal Sengupta, and Hossein Valavi

Sofia Marina, Princeton University (Spring 2026)
- Independent work for the minor in Applied and Computational Mathematics: “Evaluating Network Traffic Path Splitting Against a Transformer-Based Website Fingerprinting Attack”
- Nominated for Best Overall Project Prize among graduating seniors
- Co-advised with Maria Apostolaki

Cynthia Zhang, Princeton University (Spring 2026)
- Junior independent work: “A Tiered, Scalable, and Privacy-Preserving Stateful Firewall”
- Co-advised with Jennifer Rexford

Benjamin Herber, Princeton University (Spring 2023 - Spring 2024)
- Senior thesis and master’s research: “Tango: Secure Collaborative Route Control across the Public Internet”
- Resulted in a co-authored NSDI 2024 paper later recognized with the 2025 Applied Networking Research Prize
- Co-advised with Maria Apostolaki

Kenneth Poor, Princeton University (Summer 2022)
- Princeton/Intel REU research project as rising junior: network security, programmable-switch DDoS defenses
- Co-mentored with Jennifer Rexford

Esha Bhatia, MIT (Summer 2022)
- Princeton/Intel REU research project as rising junior: network security, programmable-switch DDoS defenses
- Co-mentored with Jennifer Rexford

Invited Talks

April 2025. “*SmartCookie: A Split-Proxy Defense on Programmable Data Planes.*” Department of Computer Science (CSC 564: Research Topics in Computer Networks). Cal Poly State University, San Luis Obispo, CA. (Virtual)

November 2024. “*On the Criticality of Integrity Protection in 5G Fronthaul Networks.*” NextG Student Research Lunch Seminar Series. Princeton University, Princeton, NJ.

October 2024. “*SmartCookie: A Split-Proxy Defense on Programmable Data Planes.*” Messaging, Malware and Mobile Anti-Abuse Working Group (M3AAWG) DDoS Special Interest Group Forum. (Virtual)

July 2024. “*Tango: Secure Collaborative Route Control across the Public Internet.*” Princeton ECE/Intel Research Experience for Undergraduate (REU) Seminar Series. Princeton University, Princeton, NJ.

January 2024. *“Getting Out of My Box: Towards Social and Intellectual Department Cohesion.”* Department of Electrical and Computer Engineering 2024 Faculty Retreat. Princeton University, Princeton, NJ.

June 2022. “SmartCookie: Distributed In-Network SYN Flooding Mitigation.” Princeton ECE/Intel Research Experience for Undergraduate (REU) Seminar Series. Princeton University, Princeton, NJ.

July 2021. “Grad School: What, Why, and How?” Princeton ECE/Intel Research Experience for Undergraduate (REU) Seminar Series. Princeton University, Princeton, NJ.

Industry Experience

Microsoft Research - Graduate Research Intern (2022-2023)
- Studied vulnerabilities in emerging 5G cellular infrastructure, discovered novel DoS attacks in O-RAN fronthaul
- Designed new countermeasures leveraging programmable data planes to defend against newly discovered attacks

NASA Jet Propulsion Laboratory - Graduate Software Computing Systems Intern (2020)
- Developed C-based emulator and flight-software tooling for the Multi-Angle Imager for Aerosols (MAIA) mission
- Tested command and telemetry processing across distributed flight-electronics infrastructure

Honeywell System Sensors - Firmware Engineer Intern (2019)
- Developed embedded firmware in C and Assembly for fire-alarm system hardware
- Redesigned circuit components of fire alarm strobe, reduced production cost of boards by over $250,000 yearly

Professional Service

Program Committee Member, ACM SIGCOMM Posters and Demos (2026)
External Reviewer and Volunteer Shepherd, IEEE Symposium on Security and Privacy (2024)
Journal Reviewer, Computer Networks (2024)
External Reviewer, USENIX Security Symposium (2023)

Community Service & Outreach

Princeton ECE Graduate Social Committee Co-Founder and Member, Princeton University (2022 - 2024)
- Proposed and established the inaugural ECE Social Committee on behalf of ECE grad students & postdocs
- Ran weekly social hangouts and larger events to strengthen department-wide community & inclusion

Princeton Lakeside Graduate Housing External Relations Delegate, Princeton University (2022 - 2024)
- Represented ~700 graduate students & families in the Lakeside community to the university
- Acted as a liaison to the Graduate School, Facilities, Housing, Transportation, Public Safety, etc.

Princeton/Intel Research Experience for Undergraduates, Princeton University (2022)
- Mentored undergraduate students in network security research, programmable switch DDoS defenses

Princeton Pre-Application Support Program, Princeton University (2021 - 2023)
- Mentored Ph.D. applicants from underrepresented backgrounds, guided their application process

Institute of Electrical & Electronics Engineers (2018 - present)
- Member, supported organization members for continual professional development

Society of Women Engineers, Temple University Chapter (2017 - 2020)
- Engaged in community and youth outreach, served on planning committee, attended national SWE conference

Joy Music Studio, Music Instructor and Executive Board Advisor (2011 - present)
- Train children and adults with tailored lesson plans for each student in violin, piano, harp, and music theory

Joyful Sound of Music, Musician and Program Coordinator (2008 - present)
- Shared music (violin, piano, harp, voice) at local elderly communities, weddings, funerals, parties, banquets